

Room-temperature quantum sensing for magnetic imaging: A Multidisciplinary Research Program

Specific Aims

Aim 1: Advance room-temperature quantum sensing for magnetic imaging via a spectroscopy-focused strategy.

Aim 2: Advance room-temperature quantum sensing for magnetic imaging via a spectroscopy-focused strategy.

Aim 3: Advance room-temperature quantum sensing for magnetic imaging via a spectroscopy-focused strategy.

Background

Rigorous, pre-registered analyses will guard against bias and support reproducibility. Our preliminary data suggest a tractable path toward measurable improvement. We will assemble a multidisciplinary team with complementary expertise. This proposal addresses room-temperature quantum sensing for magnetic imaging, a problem with substantial significance.

Approach

Prior work has established a partial understanding, yet key gaps remain. This proposal addresses room-temperature quantum sensing for magnetic imaging, a problem with substantial significance. Rigorous, pre-registered analyses will guard against bias and support reproducibility. Our preliminary data suggest a tractable path toward measurable improvement.

Innovation

We will assemble a multidisciplinary team with complementary expertise.

Investigator Context

The PI's first independent award; strong mentorship is in place. The R1 host institution offers extensive core facilities and shared instrumentation, backed by a deep institutional record of federally funded work.

Budget Justification

Total requested support is \$4,206,600 over 3 years at a R1 institution, covering personnel, materials, and dissemination.

References

1. [1] <https://doi.org/10.1016/j.cell.2016.07.054>
2. [2] <https://doi.org/10.1371/journal.pone.0173664>
3. [3] <https://doi.org/10.1145/3292500.3330701>
4. [4] <https://doi.org/10.1101/gr.229102>
5. [5] <https://doi.org/10.1056/NEJMoa2034577>

6. [6] <https://doi.org/10.1101/gr.229102>
7. [7] <https://doi.org/10.1016/j.cell.2016.07.054>
8. [8] <https://doi.org/10.1145/3292500.3330701>